2.4 Detailed specific example: Florida, USA

Florida has the second longest coastline out of all the states in the USA. However, much of it consists of sandy beaches which are at risk from erosion. Tropical storms only intensify the problem and bring other challenges.

Causes of coastal erosion

Much of Florida's landmass is composed of relatively soft sedimentary deposits, such as limestone and sandstone. These rocks are easily eroded by wave action. Florida's coastline consists mainly of beaches and lacks natural barriers, like cliffs or rocky headlands, that can help absorb wave energy and protect the shoreline from erosion.

Florida is the flattest state in the USA so is at risk of coastal erosion from flooding. Much of the state lies at or just below sea level. The highest point in Florida is only 105 metres above sea level.

Florida's location makes it particularly vulnerable to erosion from tropical storms and wave energy. The state is often in the path of tropical storms and wave energy from both the Atlantic Ocean and the Gulf of Mexico. Florida is one of the most hurricane-prone states in the USA. Tropical storms can bring intense wind and rain, as well as powerful storm surges. These storms can cause significant erosion along the coastline.

Coastal development and urbanisation in Florida have altered natural coastal processes such as longshore drift. This has reduced the ability of beaches to naturally adapt to changing conditions.

Impacts of coastal erosion

Florida has 1328 km of sandy beaches fronting the Atlantic Ocean, Straits of Florida, and the Gulf of Mexico. As of July 2023, 696 km has been critically eroded. The Florida Department of Environmental Protection (FDEP) defines critical erosion as 'a segment of the shoreline where natural processes or human activity have caused or contributed to erosion and recession of the beach or dune system to such a degree that upland development, recreational interests, wildlife habitat, or important cultural resources are threatened or lost'.

TASK 1: Study Source A

a **What is the length** of the critically eroded Fernandina Beach in km?

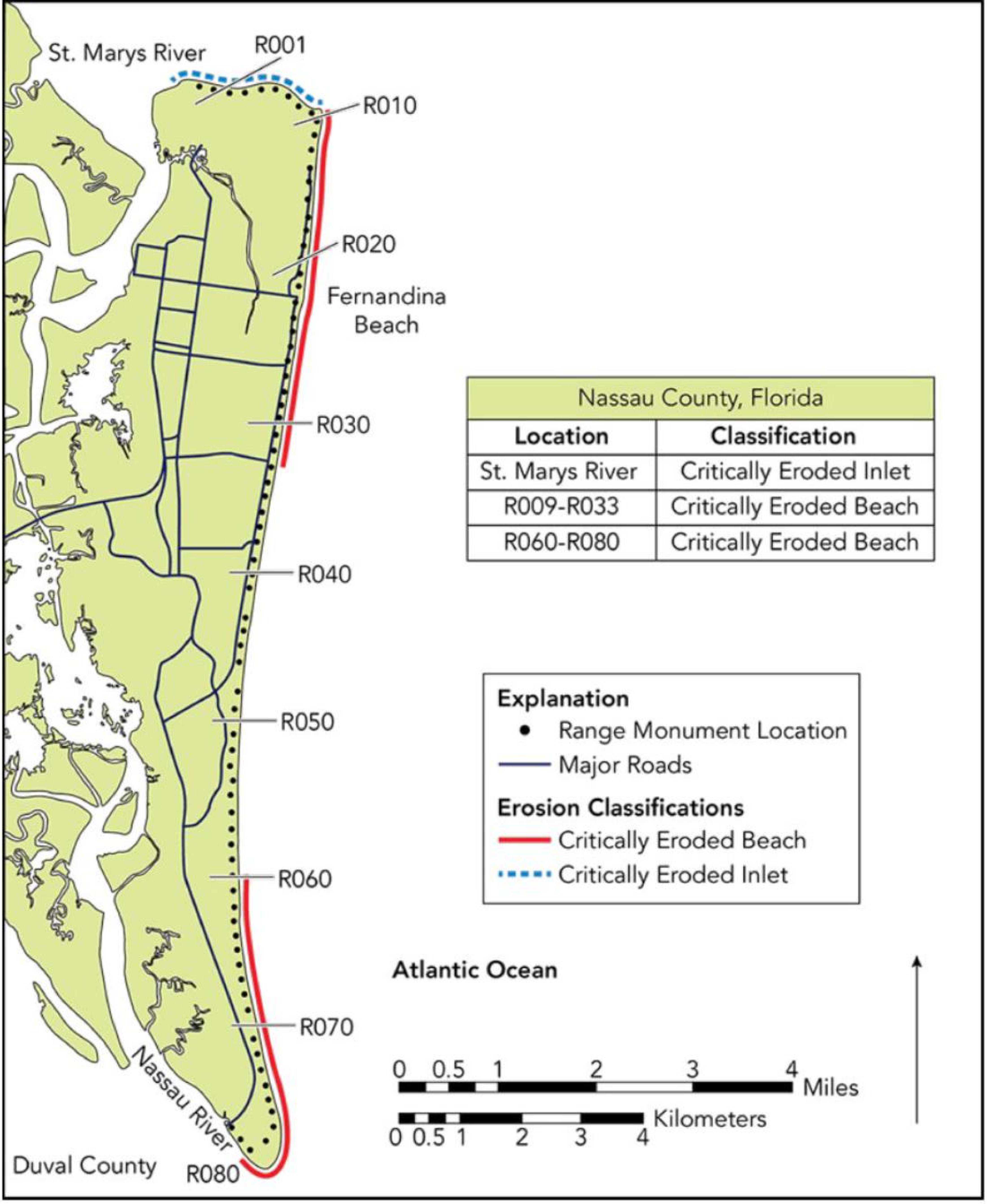
b **How does this compare** with the critically eroded beach further south near the mouth of the Nassau River, between R060 and R070?

c Which critically eroded beach poses a greater threat to infrastructure? Give evidence to support your decision.

In areas such as Miami Beach and Fort Lauderdale, ongoing coastal erosion has resulted in the gradual disappearance of sandy beaches. Erosion along Miami Beach has led to the need for extensive beach renourishment projects to maintain tourism appeal and protect infrastructure.

Coastal erosion threatens valuable coastal properties throughout the state. In places like Naples and Sarasota, erosion has caused damage to waterfront homes and hotels. Erosion weakens natural defences against storm surges, putting coastal communities at greater risk of flooding and property damage during tropical storms.

The erosion of coastal habitats is particularly evident in areas like the Everglades and Florida Keys. Mangroves, which serve as important buffers against erosion and provide critical habitat for wildlife, are increasingly vulnerable to erosion leading to habitat loss.



Source Ⓐ: Critically eroded beaches in Nassau County, Florida

Managing coastal erosion

The FDEP is responsible for developing and maintaining a long-term management plan for the restoration and maintenance of Florida's critically eroded beach fronts. This is achieved through monitoring of beach erosion and the implementation of several management techniques.

Techniques

Since 1935, Florida has spent at least $1.9 billion on beach nourishment, according to the National Beach Nourishment Database. The state government now spends between $30 and $50 million a year maintaining its beaches, and local governments contribute about the same amount.

Between 2022 and 2023, the US Army Corps of Engineers conducted beach nourishment on Miami Beach at a cost of $40.4 million. Sand was brought in by trucks from inland at a cost of $50 per cubic yard.

Groynes, breakwaters, and sea walls are all examples of hard engineering that has been installed along Florida's coastline.

Sustainable management

Soft engineering strategies, like planting vegetation, aim to work with nature to protect the coastline from erosion. In May 2022, volunteers in Virginia Key planted native trees to build the green infrastructure ahead of the 2023 hurricane season.Living shorelines are nature-based solutions that incorporate vegetation to stabilise shorelines and enhance ecosystems. In Florida, projects like the restoration of salt marshes and mangrove forests serve as living shoreline initiatives, offering both erosion control and habitat enhancement.

The Florida Coastal Management Program (FCMP) protects and enhance Florida's natural, cultural, and economic coastal resources. The programme's goal is to coordinate local, state, and federal agency activities using existing laws, to ensure that Florida's coast is as valuable to future generations as it is today.

TASK 2

Take a look at the sustainable projects that the FCMP has funded using the StoryMap on their website. Navigate to Miami Beach.

In pairs, carry out research and use your own knowledge to:

a Identify the factors contributing to coastal erosion on Miami Beach.

b Explain the impacts of coastal erosion on Miami Beach.

Source B: The beach at Sanibel Lighthouse, Florida

TASK 3: Study Source B

a Identify what management strategy is currently in place to protect the beach from erosion.

b This picture was taken before Storm Ian made landfall in September 2022, eroding the beach and damaging one leg of the lighthouse structure. In pairs, discuss what management strategies you would put in place to protect the beach from erosion in the future. Justify your decisions and be prepared to share your ideas with the class.

Managing tropical storms

Florida's Division of Emergency Management (FDEM) helps prepare communities for the impacts of tropical storms. This includes developing and updating emergency response plans, as well as conducting training exercises. Local governments in coastal areas develop evacuation plans to ensure the safety of residents and visitors during tropical storms. Evacuation zones are designated based on flood risk and storm surge potential.

Techniques

Tropical storms have significant impacts on the Florida coastline. They therefore need to be managed properly in order to reduce the impacts. Emergency shelters with medical supplies and staff are established in inland locations to provide temporary housing for evacuees during tropical storms.

Measures are taken to protect critical infrastructure such as power plants, water treatment facilities, and transportation networks from damage caused by tropical storms. This is usually in the form of hard engineering. In 2023, the Army Corps of Engineers proposed building a sea wall up to six metres high to protect Miami.

Before Hurricane Idalia (a Category four hurricane) made landfall in Keaton Beach in August 2023, warnings were issued, and Florida was put into a state of emergency. Despite sustained wind speeds of 185 km per hour and a storm surge of 2.7 metres, there were only ten fatalities. However, damage costs were estimated at around $3.6 billion.

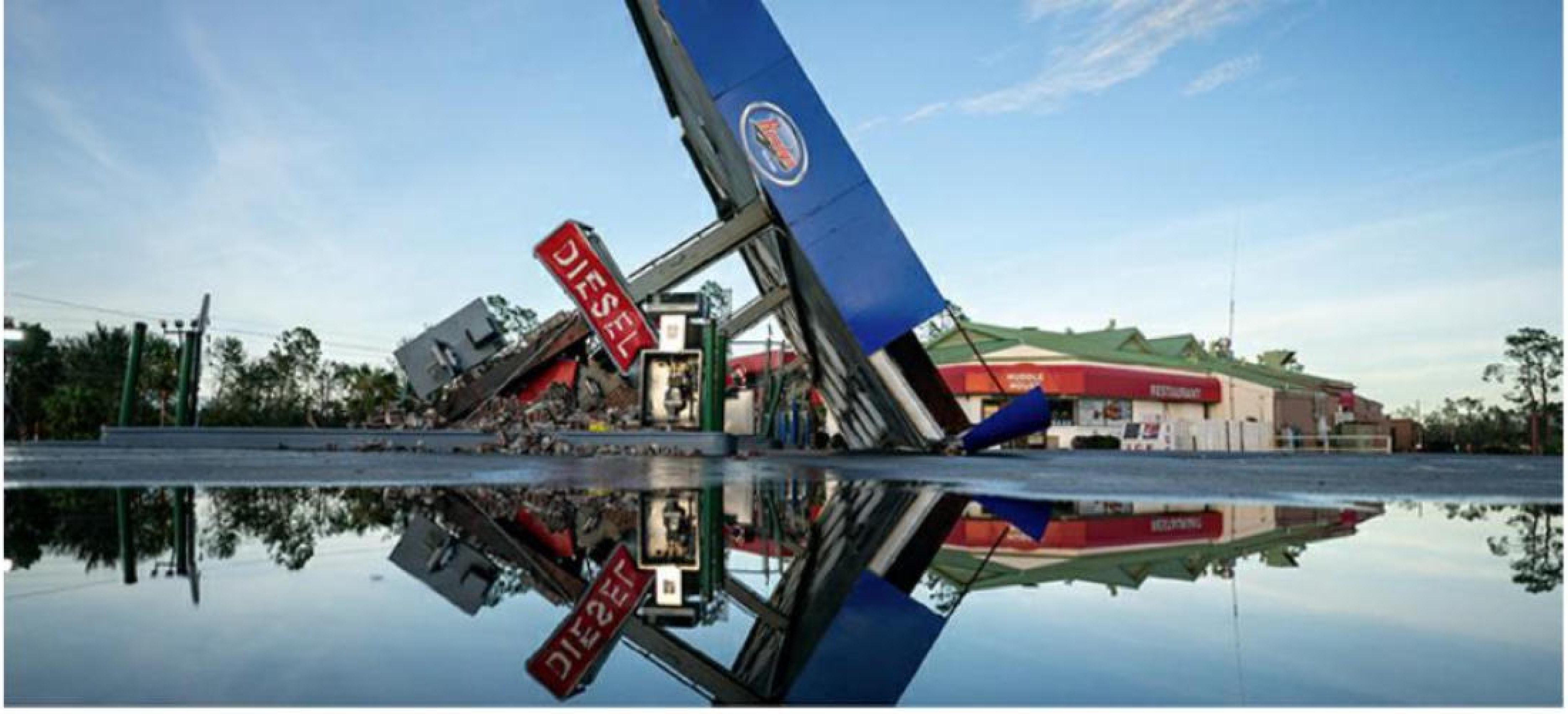


Figure 2.16: Damage caused by Hurricane Idalia, August 2023

Sustainable management

The Nature Conservancy (TNC) is a global non-profit environmental organisation based in Virginia. TNC and the City of Miami are partnering to protect Morningside Park and surrounding areas from flooding caused by rising tides and heavy rainfall due to tropical storms. The Morningside Park Coastal Resilience project, supported by the Chubb Charitable Foundation, uses nature-based solutions like mangrove restoration and elevated barriers to reduce flood risk. This initiative has not only enhanced the park's beauty, but also provides crucial protection for nearby homes and businesses during tropical storms.

TNC also collaborated with the City of Miami Beach and the Florida Power & Light Company on the Brittany Bay Park and Living Shoreline project. This project aims to mitigate the impacts of sea level rise and severe weather while also enhancing habitats for marine life. By integrating living shorelines with existing seawall improvements, the project enhances the park's resilience to high tides and storm surges. Mangroves in Florida protected approximately 625 000 people and prevented $1.5 billion in direct flood damages during Hurricane Irma in 2017.

Figure 2.17: The Brittany Bay Park and Living Shoreline help mitigate flooding from tropical storms while creating a safe environment for people and nature

TASK 4

Using examples, evaluate Florida's success at managing tropical storms.

TIP

In questions that require the use of a detailed specific example, support your ideas with evidence, such as facts and figures.